Validation and AI Use — Project 1, NVIDIA (NVDA)

Student: Drew Scheiderer · **Course:** FIN 43900, AI in Finance, Purdue, Fall 2026 **Status:** working source for `Validation-and-AI-Use.pdf`. The Locked Changed-Input Record is my Lab 11 locked prediction, human-authored with AI closed and committed before the run.

> Coursework for a learning exercise. Not investment research and not financial advice.

Decision, version, and as-of boundary

Item	Value
Decision / user	Initiate, watch/defer, or do not initiate NVDA, for a buy-side investment committee holding no position
Horizon	3–5 year holding period; five-year explicit forecast FY2027E–FY2031E plus terminal value
Repository	https://github.com/dscheid7717/SPCX-research
Analysis commit	`9222448` (engine, app, frozen output). Final frozen commit: recorded in `project-submission-manifest.csv`, column frozen_or_commit_id
Run timestamp	Frozen output generated 2026-10-15; cold-run log in `docs/cold-run-log.txt`
Source cutoffs	NVIDIA 10-K FY2026 (filed 2026-02-25) and 10-Q Q2 FY2027 (filed 2026-08-26); peer 10-Ks filed 2025-12-18 (AVGO) and 2026-02-04 (AMD); market data close 2026-10-14
Environment	Python 3.12.10; `requirements.txt` (numpy, pandas, matplotlib, scipy, yfinance, streamlit pinned ranges); finance engine is standard library only

Data and convention ledger

Material input	Source	As-of / availability	Definition / units	Transformation / exclusion	Risk
FY2026 revenue 215,938	10-K acc. 0001045810-26-000021, income statement	period 2026-01-25, filed 2026-02-25	USD M, GAAP	base year; hand-confirmed against Item 8 (Lab 10)	low
Opening balance sheet (assets 206,803, liabilities 49,510, equity 157,293)	10-K balance sheet	2026-01-25	USD M	mapped into 4 asset / 4 liability buckets; `check_opening_balance()` proves totals	low (checked)
H1 FY2027 revenue 177,837; Q2 96,221	10-Q acc. 0001045810-26-000075 (SEC XBRL `Revenues`)	period 2026-07-26, filed 2026-08-26	USD M	anchors FY2027E = H1 + 2 × Q2 = 370,279 (+71.47%)	medium: H2 is a judgment
H1 FY2027 gross margin 74.96%	10-Q (`GrossProfit` 133,299 / `Revenues`)	2026-07-26	ratio	forecast GM path starts at 75.0%	medium
FY2026 inventory write-down 4,000	10-K (XBRL `InventoryWriteDown`)	2026-01-25	USD M, in cost of revenue	normalisation: reported GM 71.07% → 72.92%	low
FY2026 equity-investment gains 8,918	10-K cash flow statement	2026-01-25	USD M, other income	excluded from forecast; stakes carried in bridge	medium
Cash + marketable securities 62,556	10-K balance sheet rows 16–17 (10,605 + 51,951)	2026-01-25	USD M	added in bridge; pro-forma holds treasury flat here	low (definition checked — see register)
Non-marketable equity securities 22,251	10-K balance sheet row 27	2026-01-25	USD M, carrying value	added in bridge (new in Edition B)	medium: carrying ≠ fair value
Debt 8,468	10-K Note 11 (999 + 7,469)	2026-01-25	USD M, carrying	subtracted in bridge; Jul-2026 debt of 33,366 disclosed, not used	medium: post-period issuance
Diluted shares 24,514	10-K EPS note	FY2026 weighted average	millions	per-share basis throughout	low (0.9% vs basic)
NVDA price $221.30	Yahoo Finance via yfinance	close 2026-10-14	USD	WACC equity weight, comps, decision	low
Risk-free 5.190%	^TNX via yfinance	close 2026-10-14	10-year UST yield	CAPM	low
Beta 2.216 (Blume 1.815)	own OLS, `nvidia/beta.py`, 5y monthly vs S&P 500	prices cached 2026-09-22	slope	CAPM; Blume in high case	**high: carries the answer**
ERP 5.0%	course convention	—	—	CAPM	high

Material input	Source	As-of / availability	Definition / units	Transformation / exclusion	Risk
Pre-tax cost of debt 2.92%	10-K Note 11, principal-weighted (Lab 06)	2026-01-25	rate	WACC (debt weight 0.16%)	negligible
Peer financials (AMD, AVGO)	10-K XBRL: AMD acc. 0000002488-26-000018; AVGO acc. 0001730168-25-000121	FY ends 2025-12-27, 2025-11-02	USD M	EV = price × diluted shares + debt − cash & ST investments; EBITDA = GAAP operating income + CF-statement D&A incl. acquisition amortisation	medium: AMD/AVGO D&A taken at XBRL tag level, not hand-read (disclosed under Limitations)
Peer prices AMD $570.82, AVGO $346.25	Yahoo Finance via yfinance	close 2026-10-14	USD	comps	low

Validation register

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
DCF engine is correct	arithmetic error in discounting / Gordon / bridge	**Known answer:** reproduce the course's 12 Lab 06 training answers ($27.50, nine grid cells, +1.78pp reverse DCF)	all 12 match to $0.005 / 0.01pp	pass	`test_project.KnownAnswerTests`
Pro-forma engine is correct	statement linkage error	**Known answer:** root Asbury training case prints published $291.75	matches	pass	`test_asbury_proforma_training_case`
Bridge cash is the right object	provider label ≠ filing line	**Independent definition check:** SEC XBRL `AvailableForSaleSecuritiesDebtSecurities` gives 39,520 for 2026-01-25; the 10-K balance sheet line "Marketable securities" gives 51,951	they differ by 12,431: the XBRL tag is the debt-securities note total, not the balance-sheet line	XBRL value **rejected**; filing line governs	10-K workbook `BALANCE_SHEET` row 17; this record
Value responds in the right direction	sign error in a driver	**Directional changed-input tests, written and committed before first execution (93a7ac9, 2026-10-15 12:39 ET):** beta ↑ → value ↓; FY2027E growth ↑ → value ↑; GM ↓ → value ↓; SBC as cash cost → value ↓; reset recovers base	all as predicted	pass	`DirectionalChangedInputTests`; `git show 93a7ac9`
Statements articulate	plugged or unrouted line	balance sheet balances every year; CFS ending cash = BS cash; base year = filed totals and revenue	max gap 0.00 on all three	pass	`ArticulationTests`; report "CHECKS"
DCF uses the pro-forma consistently	FCFF and FCFE built from different stories	FCFE = FCFF + after-tax net interest − debt repaid, every year	max gap 0.0000	pass	check "FCFF reconciles to the pro-forma's FCFE"
DCF boundaries	value moves the wrong way, or Gordon used where undefined	grid strictly ↓ in WACC and ↑ in g over 5 × 5; g ≥ WACC refuses	monotonic; refusal returns no value	pass	`DcfBoundaryTests`
Bridge reconciles	EV-to-equity arithmetic	EV + 62,556 + 22,251 − 8,468 = equity; × shares both ways	exact	pass	`BridgeTests`
Peer policy applied consistently	definition drift between companies	every company carries every field; fiscal ends within 120 days; positive EBITDA/EPS; EV/EBITDA recomputes; NVDA recomputed by hand	all pass; NVDA 40.3× by hand	pass	`PeerPolicyTests`
Highest-risk failure (model)	a broken sheet is valued silently	inject SBC-without-equity-credit and an opening mis-map; model must refuse, gap must equal SBC exactly; next run must be clean	refuses; gap = SBC; no leak	pass	`HighestRiskFailureTests`; app fault switch
Highest-risk failure (evidence)	a stale driver carried forward	compare Lab 10 FY2027E growth (30%) to the 10-Q filed 2026-08-26	H1 already +95.8% YoY; 30% would need H2 revenue to fall to 102,882	**Lab 10 driver failed**; replaced (Edition B)	`analysis.py` DRIVER_RECORD

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
Decision rule can be moved by evidence	rule that no downside can alter	first-run review: does any downside case change the action?	**failed** under the median-only rule (`93a7ac9`); after adding the earnings-persistence leg and my Lab 12 67% margin floor, the low case, hyperscaler-pause and margin-compression what-ifs give DO NOT INITIATE	revised, disclosed	`9222448` commit message; `analysis.py` decision-rule block
Frozen output is reproducible	hand-edited or stale visible output	fresh `run_analysis()` equals `visible_output.json`	equal	pass	`ReproducibilityTests`
Clean environment reproduces	hidden local state	**cold run:** fresh clone, fresh venv, `pip install -r requirements.txt`, tests, analysis, diff against frozen report, app smoke test	run 1 **failed** at the app step (`streamlit.proto` missing: Windows long paths disabled, 176-character folder prefix); run 2 from a short path clean — 27/27 tests, report identical to frozen, app WATCH/DEFER $78.83, 291 s	failure diagnosed as environment, not code; README now tells Windows users to clone to a short path	`docs/cold-run-log.txt`

Locked Changed-Input Record — human prediction first; AI closed during execution

This record is my **Lab 11 locked prediction, taken as is**. It was human-authored with AI closed and committed before any sensitivity run, as Lab 11 required. Full text: `nvidia/NVIDIA_2026-09-29_lab11_locked_prediction.md`. It was run on the Lab 10 pro-forma, the same engine `nvidia/analysis.py` now drives.

Precommit before the run and before AI assistance on the prediction

Precommit timestamp	Repository commit	Material input	Old → new value/units	Expected output direction	Expected decision effect
2026-09-29 15:34:46 −04:00	`dbf76fd` ("Lock the Lab 11 sensitivity prediction before any run")	(1) `REVENUE_GROWTH`; (2) `GROSS_MARGIN` — in `nvidia/proforma.py`	(1) 0.30/0.22/0.16/0.11/0.07 → uniform ±0.02 (±2pp, decimal) every year; (2) 0.710→0.690 path → uniform ±0.02 (±2pp, not ±2%)	Both move value per share, FY2031E operating profit and FCFE in the **same direction as the input**. Growth: **1–4% of value per pp**. Margin: **≤1% per pp**. Growth's span > margin's; span ratio ~1×–4×	Not a separate field in the Lab 11 template, so not recorded before the run. The precommitted words were directional only: lower growth → value "would drop significantly"; lower margin → value drops, "not to as large of a magnitude"

Preserved execution result

Before/after output	Actual decision effect	Prediction reconciliation	Failure diagnosis or why action did not change	Evidence path
Base $58.23. Growth −2/+2pp: $54.51 / $62.17 (span 7.66; 3.29% of base per pp). Margin −2/+2pp: $56.12 / $60.33 (span 4.21; 1.81% per pp)	**None — WATCH–DEFER unchanged**	4 of 5 claims correct: ranking, sign, growth size, span ratio (1.8×). **Wrong:** margin size, 1.81% per pp vs ≤1% predicted (~1.8× my ceiling)	*Failure:* (a) a point of margin is a point of **revenue** (471,842 in FY2031E), so 2pp = 9,437 of gross profit vs 255,618 operating profit; (b) terminal value is 59% of value and capitalises the final year, amplifying both drivers. One stated reason (strategic capital) is not a mechanism the model contains. *No change because:* both drivers together span $54.51–$62.17 against a $224.58 price; no operating assumption defensible from NVIDIA's history closes a $166 gap — the disagreement is the discount rate	`nvidia/NVIDIA_2026-09-29_lab11_locked_prediction.md`; `nvidia/output/NVIDIA_2026-09-29_lab11_sensitivity.txt`; `git show dbf76fd`

Post-run edits, declared in the record itself: after the run I filled two per-driver tables that were empty scaffolding in `dbf76fd`; the quantified forecast was already in the committed text and `git diff dbf76fd HEAD` on that file shows it unchanged.

Carried to the Project 1 model (AI-run, 2026-10-15; not part of the locked record): the same ±2pp shifts on the Edition B base give growth $74.27 / $78.83 / $83.66 (2.98% per pp) and margin $76.10 / $78.83 / $81.57 (1.73% per pp). The Lab 11 ranking holds; the margin under-prediction persists at the same size; the action is WATCH/DEFER throughout.

Judgment carry-forward — Labs 06–12 into Project 1

My lab judgments are the default. Where Project 1 changes one, the reason is stated.

Judgment	Lab source	Lab value	Project 1 value	Kept / changed	Why
FY2027E revenue growth	Lab 06, 10	30%	**71.5%** (H1 reported + H2 at Q2 run rate)	**changed**	The Q2 FY2027 10-Q (filed 2026-08-26) shows H1 +95.8%; 30% would require H2 revenue to *fall* to 102,882
FY2028E–31E growth	Lab 06, 10	22 / 16 / 11 / 7%	22 / 16 / 11 / 7%	kept	No evidence against the decay path
Gross margin path	Lab 10	71.0% → 69.0%	**75.0%** → 69.0%	start changed, endpoint kept	H1 FY2027 74.96%; FY2026's 71.07% was depressed by a 4,000 write-down (normalised 72.92%). The competitive-fade reasoning and 69% endpoint stand
Gross margin floor	Lab 11 range, Lab 12 trigger	67% → do not initiate	67% → do not initiate, **now enforced by the decision rule**	kept, strengthened	The custom-silicon what-if (65%) now flips to DO NOT INITIATE, as my Lab 12 trigger says it should
R&D / revenue	Lab 10	8.5% → 10.5%, rising	**7.5%** → 10.5%	start changed, logic kept	H1 FY2027 7.52%; the ratio still rises as growth slows
SG&A / gross profit	Lab 10	3.0% flat	**2.0%** → 3.0% by FY2029E	start changed	H1 FY2027 1.99%; returns to my 3.0%
Tax rate	Lab 10	16%	16%	kept	H1 FY2027 effective 16.5% supports it
Capex, inventory days, SBC, D&A, supply-obligation ratios	Lab 10	FY2026 history	same	kept	—
Treasury held flat; surplus distributed	Lab 10 correction	62,556	same	kept	Prevents double-counting interest on distributed cash
Buybacks / dividends	Lab 10	40,000 / 1,000	**78,000 / 12,600**	changed	H1 FY2027 annualised (39,044 / 6,290). No value effect: the surplus distribution is a residual
Beta	Lab 09, 10	2.216 base; Blume 1.815 "as a sensitivity, not buried"	same; Blume in the high case and a what-if	kept	—
Risk-free rate / price	Lab 06, 10	4.944%, $218.36 (09-10); $224.58 (09-24)	**5.190%, $221.30 (10-14)**	changed	One date for every market fact, including peers
ERP, terminal growth, diluted shares	Lab 06	5%, 3%, 24,514M	same	kept	—
Bridge	Lab 06	EV + 62,556 − 8,468	**+ 22,251 non-marketable equity**	changed	My Lab 11 note: those stakes are excluded from the forecast "by construction". Left out of the bridge too, their value would vanish
Peer set and policy	Lab 08	AMD, AVGO; INTC excluded; one price date; GAAP	same; **EV/EBITDA added**	kept + extended	Project requires an EV multiple; EBITDA neutralises the acquisition amortisation Lab 08 found distorting AMD's P/E
Never average the methods	Lab 08, 12	—	same	kept	—
Call	Labs 08–12	WATCH–DEFER	WATCH/DEFER	kept	—
Initiate trigger	Lab 06, 08, 12	price below ~$114 (0.5× edge of the reasonableness band)	**price ≤ base-case value ($79)**	**changed**	The 0.5×–2.0× band tests whether a model is sane against the market; it is not a margin of safety. $114 meant paying twice my own value. (It sits close to the new high case, $115.)
Do-not-initiate: segment growth	Lab 12	Compute & Networking slower than my 30% FY2027 path	YoY revenue decline → DO NOT INITIATE; C&N growth below path → escalate and rerun	changed	The 30% path was contradicted by the 10-Q
Do-not-initiate: supply obligations	Lab 12	ratio above 12.8% of inventory	above 12.8% → **escalate and rerun**	threshold kept, action changed	12.8% is today's level, so any uptick would flip the call; one quarter's accrual timing is a reason to rerun, not to decide
Re-underwrite conditions	Lab 12	defensible Ke materially below 16%; peers at 20–30× trailing GAAP P/E	same	kept	—

Judgment	Lab source	Lab value	Project 1 value	Kept / changed	Why
Research priority	Lab 11, 12	beta first, C&N growth second	same	kept	Reverse DCF: the price needs beta 0.51

Named AI-use register

Tool / model and version	Date	Material task	Output / claim used or considered	Independent check	Disposition	Effect on decision
ChatGPT / Codex; Claude Code	2026-09-08 → 2026-10-01	Lab 06–12 drafting and code, as disclosed in each lab write-up	Lab models and write-ups	known-answer reproduction each lab; hand checks recorded in the labs	accepted with the checks recorded there	built the Lab 06–11 evidence base
Claude Code (Claude Opus 5.5)	2026-10-15	Pull NVDA and peer filing facts from SEC XBRL company-facts	candidate value 39,520 for "marketable securities" (`AvailableForSaleSecuritiesDebtSecurities`)	10-K balance sheet, workbook BALANCE_SHEET row 17: 51,951	**rejected** — wrong object (note total, not the balance-sheet line)	none: bridge kept at 62,556
Claude Code (Claude Opus 5.5)	2026-10-15	Review of Lab 10 drivers against the Q2 FY2027 10-Q	claim that FY2027E growth of 30% is contradicted by H1 +95.8%	quarterly facts sum to the half-years: 81,615 + 96,221 = 177,836 (vs 177,837, $1M rounding); 44,062 + 46,743 = 90,805	**accepted**; Lab 10 driver **modified** (see carry-forward)	base value $58 → $79; action unchanged
Claude Code (Claude Opus 5.5)	2026-10-15	Proposed Edition B driver values, low/high cases, what-ifs	values in `analysis.py` BASE_INPUTS, CASES, WHAT_IFS	each anchored to a cited 10-Q / 10-K figure; lab judgments kept unless the carry-forward table states a reason	**modified**: lab judgments kept as default; changes limited to those in the carry-forward table	sets the $49–$115 range
Claude Code (Claude Opus 5.5)	2026-10-15	Decision rule	first version could not be moved by any downside case	first run of the engine	**modified**: earnings-persistence leg (9222448), then my Lab 12 67% margin floor	low case, hyperscaler pause and margin compression flip to DO NOT INITIATE
Claude Code (Claude Opus 5.5)	2026-10-15	Wrote `analysis.py`, `app.py`, `test_project.py`	engine, app and 27 tests	12 course known answers + Asbury $291.75 reproduced; articulation checks; two cold runs	**accepted** after tests and cold run	delivery
Claude Code subagent	2026-10-15	Read the public course repo for Project 1 requirements	"the analytic rubric file is not in the public repo"	repository tree listing	**qualified**: true of the public repo only; Brightspace governs	none

Limitations, monitoring, and kill/escalation rules

Limitations

- The forecast opens on the FY2026 balance sheet (2026-01-25) and discounts from that date; the price is 2026-10-14. Rolling the base value forward roughly 0.7 year at WACC would raise it about 11% before deducting H1 distributions — far from closing the gap to price.
- WACC rests almost entirely on beta. Raw 2.216 vs Blume 1.815 is a ~200bp swing ($79 → $93).
- The reverse DCF puts the disagreement in the discount rate: the price needs a beta of 0.51 (cost of equity 7.74%) on my operating forecast, or FY2028–31 growth 45pp above my path.
- Two peers only; trailing multiples lag NVIDIA's growth; AMD's multiple is inflated by thin margins and acquisition amortisation.
- AMD and Broadcom D&A are taken from SEC XBRL tags, not hand-read from each cash flow statement.
- Scenario endpoints are judgments, not probabilities.

Monitoring and kill/escalation rules

Trigger	Metric and threshold	Window	Action
Initiate	NVDA close ≤ base-case value (~$79)	daily	escalate to committee for initiation
Demand turn	any 10-Q with YoY revenue decline	each 10-Q (next: Q3 FY2027)	DO NOT INITIATE (low-case path)
Competitive entry	gross margin below 67% (Lab 11/12 floor)	each 10-Q	DO NOT INITIATE
Relative value lost	NVDA trailing EV/EBITDA ≥ lowest admitted peer	each 10-K / price review	DO NOT INITIATE
Over-ordering	excess purchase obligations > 12.8% of inventory (Lab 12)	each 10-Q	escalate and rerun
Segment deceleration	Compute & Networking growth below the forecast path	each 10-Q	escalate and rerun
Re-underwrite	defensible Ke materially below 16%; or peers at 20–30× trailing GAAP P/E (Lab 12)	as evidence arrives	rerun valuation
Model integrity	any check fails	every run	kill: no value is reported until fixed